

Demonstrating Myo Action: Accelerating Voluntary Actions via Electromyography and Muscle Stimulation

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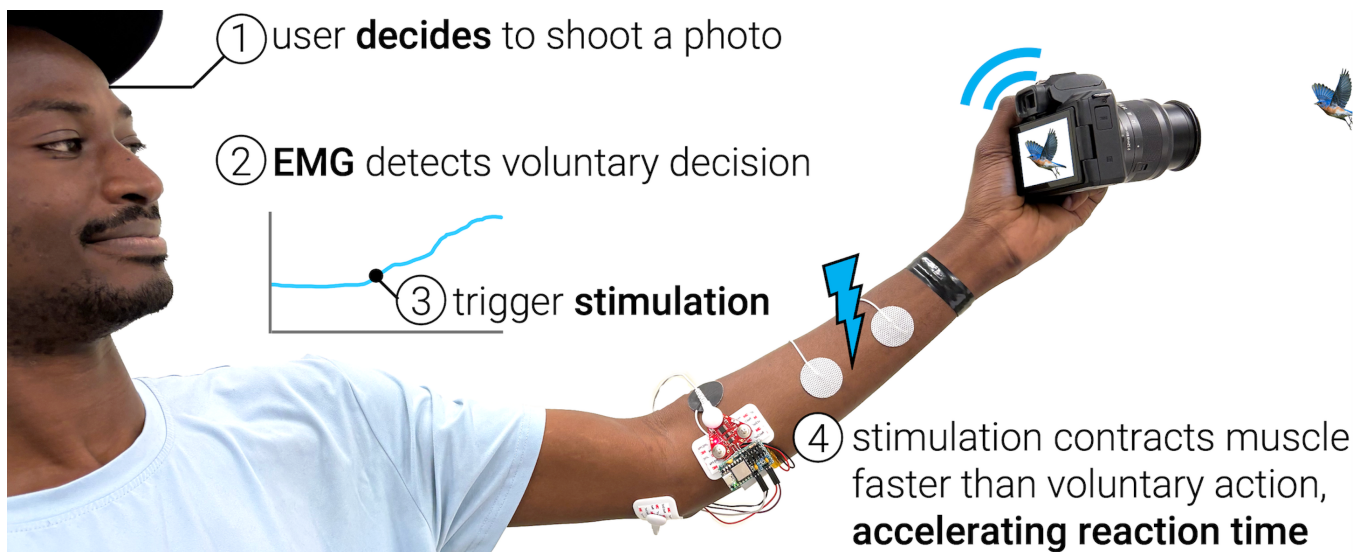


Figure 1: We present Myo-Action, an interaction technique to accelerate users' voluntary action without overriding their intentions. (a1, a2) Our device detects the user's intention to execute a physical action via electromyography (EMG) and (a3) activates electrical muscle stimulation (EMS) at an early onset of EMG. (a4) The key is that EMS takes less time to start contracting muscles compared to typical voluntary actions.

Abstract

We propose a technique for accelerating users' action without overriding intention, thereby preserving agency. In our approach, it is the user's muscle signals, detected via electromyography (EMG), that trigger electrical-muscle-stimulation (EMS) without external sensors or stimulation-timing calibration. The key to enable this "agentic speedup" is a synergy between EMG and EMS: EMG can detect an early-onset of the neural-response; EMS can contract a muscle faster than a typical voluntary-contraction. This—coupled with our low-latency system ($\sim 290 \mu s$)—results in an accelerated

reaction-time, even though the haptic-assistance is initiated after the muscle-signal. Our novel approach: (1) accelerated users' reaction-time by ~ 23 ms compared to voluntary-action; (2) preserved agency in decision-involving actions (i.e., go/no-go trials), which existing muscle-stimulation techniques cannot achieve.

CCS Concepts

• Human-centered computing → Interaction techniques.

Keywords

Agency, Electrical Muscle Stimulation, Electromyography, Intention

ACM Reference Format:

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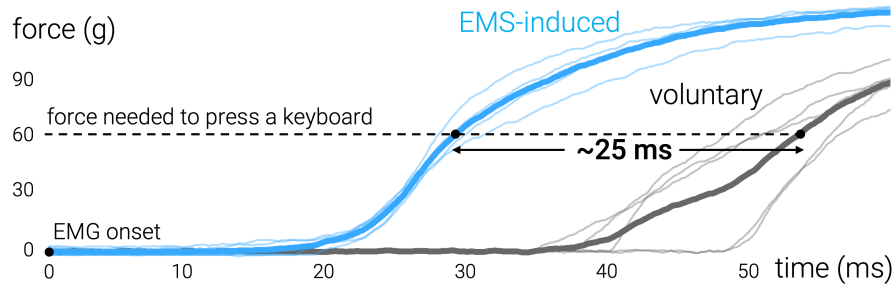


Figure 2: An observation of our key principle: EMS-induced contractions reached the force-level required to register an input ~25 ms faster than voluntary contraction. Example data from one user is shown (averaged over five trials per condition). Measurements were based on the same force-sensor, EMS stimulator, and oscilloscope employed in our latency evaluation.

1 Introduction

Accelerating movements via electrical-muscle-stimulation (EMS) can enhance users’ reaction-times in a wide range of physical tasks from high-speed photography [13]; grabbing a falling object before it hits the ground [8, 14]; playing e-sports [19]; and even steering users to safety [15]. These accelerations can even train users [9].

Unfortunately, this approach also deprives users of their sense-of-agency (the feeling of “I did this”) by overriding their intention to move—these interfaces do not allow users to opt-out from the EMS-movement.

While recent work has mitigated some loss of agency by fine-tuning the timing of EMS [8, 16], this only makes so-called preemptive accelerations agentic when they are closely-aligned with the user’s own timing (demanding a lengthy calibration). However, these techniques fail if the user does not decide to move, in which the system delivers the preemptive stimulation and the user is left with a drastic sensory conflict. Prior work has measured this drop in agency (system does not guess user’s intention) and found that it is even more detrimental [16].

While researchers have investigated closed-loop body-actuation systems with various sensing techniques, there has not yet been a successful closed-loop system that accelerates reaction-time. Thus far, the most prevalent sensing modality for closed-loop EMS is motion tracking (e.g., accelerometers, optical marker-based systems). However, such approaches only capture movements that are already underway and are unable to detect the cues preceding an intended movement—required for accelerating a user’s physical action. By contrast, it is possible to detect cues that precede a movement by detecting neural-responses (e.g., brain-sensing via EEG or muscle-sensing via EMG).

In this paper, we introduce Myo-Action, a novel approach that accelerates users’ movements while preserving their sense-of-agency (Figure 1). Unlike prior techniques, our method leverages the user’s neural-signals as the trigger for assistance. Specifically, we use electromyography (EMG) to detect the earliest onset of muscle activation—indicating that the user has already committed to move—and immediately apply EMS to the same muscle. Because EMS induces a contraction faster than voluntary-activation (and our system-latency is only ~290 μ s), this results in a measurable reduction in reaction-time. We demonstrate that this approach accelerates reaction-times by approximately 23 ms compared to unassisted actions, preserves agency even in decision-based tasks, and

is experienced positively in real-world scenarios, thus establishing Myo-Action as a new direction for haptic-assistance [17].

2 Myo-Action: Principle of Operation

Our technique, Myo-Action¹, preserves users’ agency while accelerating movement by detecting neural activity of a specific muscle via EMG and triggering EMS. Although this may seem paradoxical—i.e., how could a system speed up a contraction that is already ongoing?—we exploit two key principles: (1) EMG detects neural activation in a target muscle prior to any muscular displacement; and (2) EMS can induce faster than voluntary muscle contraction.

EMG detects neural activation in a target muscle before movement. The first principle that enables our approach is that EMG can detect a muscle’s neural activity, preceding measurable movement or force. This time window, known as *electromechanical delay* [3, 4], is caused by excitation-contraction coupling and tendon and joint compliance and is typically tens of milliseconds. It is important to note that EMG must target a specific muscle with a pair of electrodes placed over the muscle belly [12]. In principle, this underlying per-muscle selectivity reduces interference from other muscles during detection in multi-muscle movements [5]—though in this work, we limit our device to work in simple movements and have not characterized multi-muscle movements.

EMS causes faster muscle contraction than voluntary control. Figure 2 depicts how an electrically-induced contraction tends to outperform a voluntary contraction. This is in line with findings in neuroscience: “(...) delay was greater in the voluntary condition (22.8 +/- 8.2 ms) compared to the involuntary condition (9.7 +/- 3.1 ms; $p < .001$)” [7] and “delay of the involuntary contractions (...) was significantly shorter than that of the voluntary contractions [(...) $P < 0.05$]” [18]. The process that best explains this is from how our nervous system chooses which muscle fibers to use for a movement. This process, known as motor-unit recruitment, follows *Henneman’s size principle* (1965): “motor neurons with large cell bodies tend to innervate fast-twitch, high-force, less fatigue-resistant muscle fibers, whereas motor neurons with small cell bodies tend to innervate slow-twitch, low-force, fatigue-resistant muscle fibers” [2, 6]. This allows users to control forces precisely—if motor-units that can only control large-forces would be recruited, human-level dexterity would be challenging. Turning back to the

¹We named it **Myo-Action** since it uses **My**ography (EMG) and its accelerated-movements are felt as “**My Own**” Action.

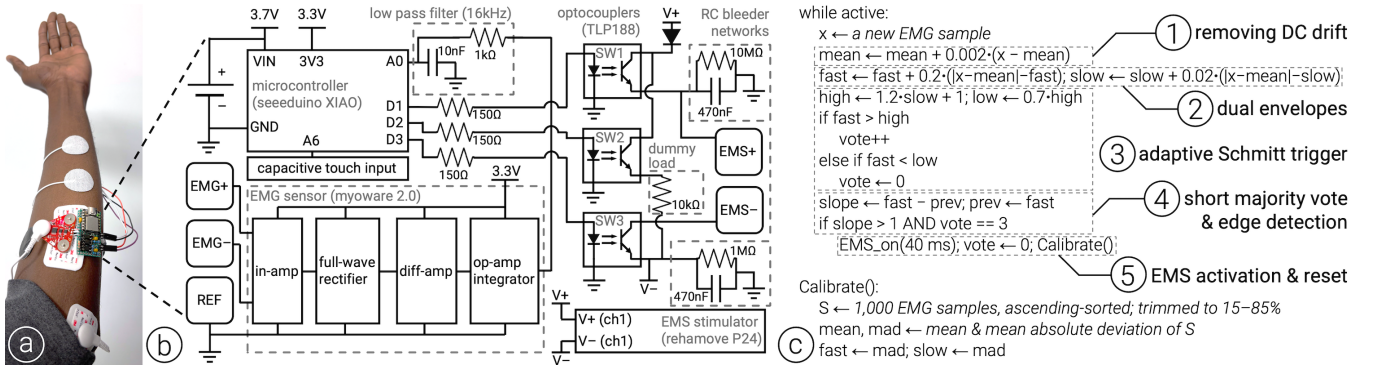


Figure 3: (a) Our hardware system. (b) Circuit design. (c) Firmware-level signal processing to further mitigate false triggering.

fundamentals of electrical-stimulation, we can deduce the effect of EMS on motor-unit recruiting. Since any type of electrical stimulation is non-selective [11], EMS will recruit both types of fibers concurrently [7]. This results in large motor-units being stimulated, causing “fast-twitch, high-force” movements [7] which accelerate the initial-phase of the contraction [3, 4].

3 Implementation: Engineering Fast EMG Detection to Speed up EMS Response

Hardware overview. As shown in Figure 3 (a), our device is comprised of: EMG-amplifier (Myoware-2.0), microcontroller, optical-switching stage, analog-filtering stage, and a medical-grade EMS-stimulator (Rehamove-P24). Electrodes are placed atop a muscle to cause movement. The EMG sensor is also attached atop the same muscle via another pair of electrodes plus a reference. We achieve low-latency by having the stimulator continuously output a pulse-train (800- μ s pulse-width, 400Hz) and gating the stimulation via optical-switches upon detecting EMG-signals.

Circuit design. Two circuits enable our device (Figure 3b): (1) an optical-switching stage that gates the stimulator output to the user upon EMG detection; and (2) an analog-filtering stage that stabilizes the EMG-sensing by mitigating both background and EMS-induced noise. First, the optical-switching stage uses three optocouplers (TLP188; 350V rating; $\sim 1\text{-}\mu$ s turn-on), driven by a SAMD21’s digital-outputs. When EMG is not detected, SW1 and SW3 are OFF and SW2 is ON, routing the EMS to a dummy load to suppress residual noise. Upon detection, SW1 and SW3 turn ON while SW2 turns OFF for 40 ms, delivering stimulation to the user.

Second, the analog-filtering stage employs RC bleeder networks and a 16kHz low-pass filter. During EMS, the bleeder networks mitigate leakage currents to the sensor-side. The low-pass smooths high-frequency noise at the ADC input, enabling cleaner sampling of the EMG amplified-output.

User input to activate system. Finally, users can toggle the system to enable or disable EMG triggering via a capacitive-touch button (SAMD21 QTouch), allowing them to control when EMG is monitored, potentially reducing unintended stimulation in more mobile or less-controlled activities or during prolonged use.

Microcontroller’s signal-processing. Our detection-algorithm runs on the microcontroller to improve the reliability of the EMS-triggering (Figure 3c). When the detection system is activated (button press by user), it runs a calibration, which acquires 1000 EMG samples as a baseline of the noise-level; these values seed the filters. During operation, DC-drift removal subtracts a slow baseline from each incoming sample, so only rapid muscle activity remains. A dual-envelope then runs at two timescales: a fast envelope that follows bursts with microsecond-order latency; and a slow envelope that adapts to the ambient noise-floor/EMS-induced-artifacts. From these, we form an adaptive Schmitt trigger with dynamic upper/lower thresholds and hysteresis, preventing chatter near boundaries. Finally, a short majority vote and edge-check require three consecutive hits and a rising slope, rejecting isolated spikes. When satisfied, we gate the stimulator output for 40 ms and then schedule a recalibration to update the noise baseline.

Latency. We evaluated this with a 100MHz-oscilloscope (GWInstek GDS-1102), probing signal input+output. We measured ~ 115



Figure 4: Our envisioned eSports application.

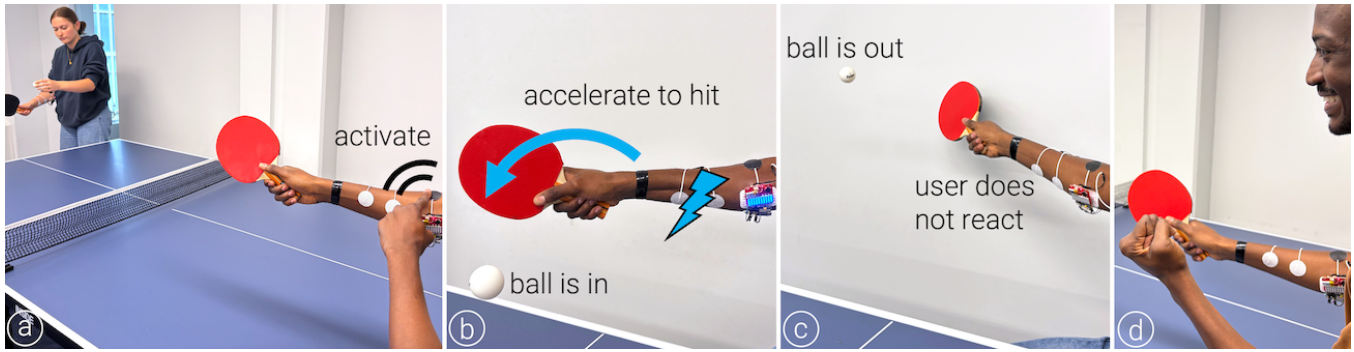


Figure 5: Our envisioned ping-pong application.

μs through the EMG-amplifier and $\sim 175 \mu\text{s}$ through our analog-filtering and optical-switching stages. Thus, latency from EMG-detection to EMS-activation is $\sim 290 \mu\text{s}$.

Safety. Our medical-grade stimulator controls electrical currents independent of skin-impedance, and our software limits the maximum output to 20 mA—25 times below the safety limit [10].

4 Envisioned Applications

We envision applications that illustrate more diverse uses of Myo-Action. (for the sake of visual clarity, we added a Myoware-LED-shield [1] showing real-time EMG activity). Note that our controlled experiment focused only on a simple movement. These envisioned applications do not intend to supplant evaluations or the novel algorithms that might generalize to more complex movements.

4.1 eSports

The 20-ms speedup in reaction-time might impact video-gaming performance (Figure 4). When gaining a power-up, the player activates Myo-Action to initiate the detection thread (Figure 4a). When they intend to use the power-up, our device detects the EMG onset, enabling its faster activation—and leaving the opponent less time to dodge (Figure 4b-c).

4.2 Performance Augmentation (e.g., PingPong)

Myo-Action might augment users' performance in sports, e.g., ping-pong (Figure 5). For instance, our device could accelerate the movement to hit the ball when the user intends to (i.e., when the ball is in), whereas when they do not intend to hit it (i.e., when the ball is

out), it respects their intention, causing no involuntary movement (Figure 5b-c). Note that follow-up research would be needed to ensure Myo-Action's stability in such unconstrained use cases.

4.3 Accelerating Multiple Muscles (e.g., Bimanual Interactions)

We can extend Myo-Action to multiple muscles, e.g., in bimanual interactions (Figure 6). A user might wear two devices linked to a single activation button, allowing the system to accelerate steering in a driving-simulator, according to their intended way to turn. Again, further research is needed to ensure sensing-stability in such unconstrained use cases.

5 Conclusion

In this paper, we demonstrated a new concept for accelerating users' action without overriding their intentionality—users have agency to decide if they want to move, not the haptic interface. Our approach combines EMG and EMS to realize this agentic speedup.

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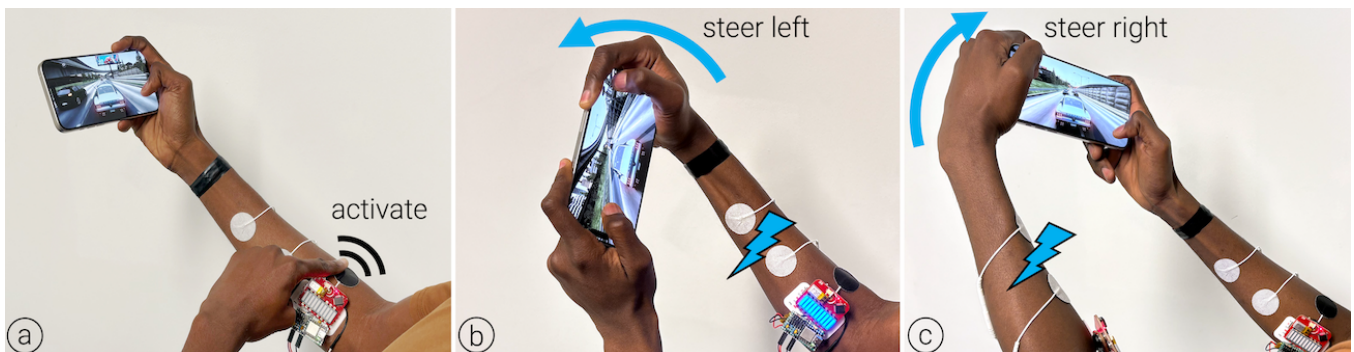


Figure 6: Our envisioned bimanual-interaction (driving-simulator) application.

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